



第一章 集合习题

习题 1.1 (Exercise 1.1)

1. 用符号 “ $\in$ ” 或 “ $\notin$ ” 填空 . (Fill in the blanks with the symbols “ $\in$ ” or “ $\notin$ ” .)

(1) $5 \underline{\hspace{1cm}} \mathbb{N}$, $-3 \underline{\hspace{1cm}} \mathbb{N}$, $5.3 \underline{\hspace{1cm}} \mathbb{N}$, $0.25 \underline{\hspace{1cm}} \mathbb{N}$, $\frac{\sqrt{2}}{3} \underline{\hspace{1cm}} \mathbb{N}$.

(2) $8 \underline{\hspace{1cm}} \mathbb{Z}$, $\frac{3}{2} \underline{\hspace{1cm}} \mathbb{Z}$, $3 \underline{\hspace{1cm}} \mathbb{Z}$, $-8 \underline{\hspace{1cm}} \mathbb{Z}$, $\sqrt{4} \underline{\hspace{1cm}} \mathbb{Z}$, $\sqrt{3} \underline{\hspace{1cm}} \mathbb{Z}$, $(\sqrt{5})^2 \underline{\hspace{1cm}} \mathbb{Z}$, $\frac{\pi}{3} \underline{\hspace{1cm}} \mathbb{Z}$.

(3) $3.14 \underline{\hspace{1cm}} \mathbb{Q}$, $\frac{2}{5} \underline{\hspace{1cm}} \mathbb{Q}$, $-\frac{7}{4} \underline{\hspace{1cm}} \mathbb{Q}$, $3^0 \underline{\hspace{1cm}} \mathbb{Q}$, $-7 \underline{\hspace{1cm}} \mathbb{Q}$, $\sqrt{16} \underline{\hspace{1cm}} \mathbb{Q}$, $\sqrt{3} \underline{\hspace{1cm}} \mathbb{Q}$, $(\sqrt{3})^2 \underline{\hspace{1cm}} \mathbb{Q}$, $-3\pi \underline{\hspace{1cm}} \mathbb{Q}$.

(4) $2 \underline{\hspace{1cm}} \mathbb{R}$, $\frac{2}{5} \underline{\hspace{1cm}} \mathbb{R}$, $-\frac{8}{3} \underline{\hspace{1cm}} \mathbb{R}$, $0.56 \underline{\hspace{1cm}} \mathbb{R}$, $-5 \underline{\hspace{1cm}} \mathbb{R}$, $-\sqrt{4} \underline{\hspace{1cm}} \mathbb{R}$, $\sqrt{5} \underline{\hspace{1cm}} \mathbb{R}$, $(\sqrt{7})^2 \underline{\hspace{1cm}} \mathbb{R}$, $\frac{\pi}{5} \underline{\hspace{1cm}} \mathbb{R}$.

2. 设集合 $A = \{x \mid -5 < x \leq 5\}$, 则下列关系正确的是 () . (Let set $A = \{x \mid -5 < x \leq 5\}$, then which of the following relationships is correct () .)

A. $\frac{5}{3} \notin A$ B. $-\frac{7}{3} \notin A$ C. $4.3 \in A$ D. $-5 \in A$

3. 设集合 $B = \{x \in \mathbb{Z} \mid -4 \leq x < 2\}$, 则下列关系正确的是 () . (Let set $B = \{x \in \mathbb{Z} \mid -4 \leq x < 2\}$, then which of the following relationships is correct () .)

A. $-4 \notin B$ B. $-2.5 \notin B$ C. $2 \in B$ D. $1.5 \in B$

4. 集合 $\{x \in \mathbb{Z}^+ \mid x \leq 4\} = ()$. (Set $\{x \in \mathbb{Z}^+ \mid x \leq 4\} = ()$.)

A. $\{0, 1, 2, 3, 4\}$ B. $\{1, 2, 3, 4\}$ C. $\{0, 1, 2, 3, 4, 5\}$ D. $\{1, 2, 3, 4, 5\}$

5. 下列说法正确的是 () . (Which of the following statements is correct () .)

A. If $a \in \mathbb{N}$, $b \in \mathbb{N}$, then $a - b \in \mathbb{N}$ B. If $a \in \mathbb{Z}^+$, then $a \in \mathbb{Q}$

C. If $a \geq 0$, then $a \in \mathbb{N}$ D. If $a \in \mathbb{Z}$, then $a \notin \mathbb{Q}$

习题 1.2 (Exercise 1.2)

1. 用符号 “ $\in$ ” “ $\notin$ ” “ $\subseteq$ ” “ $\subsetneq$ ” “ $\not\subseteq$ ” 或 “ $=$ ” 填空. (Fill in the blanks with the symbols

“ $\in$ ” “ $\notin$ ” “ $\subseteq$ ” “ $\subsetneq$ ” “ $\not\subseteq$ ” or “ $=$ ”.)

(1) $3 \underline{\hspace{1cm}} \{-3, -2, 1, 2, 3\};$ (1) $3 \underline{\hspace{1cm}} \{-3, -2, 1, 2, 3\};$

(3) $\{x \mid -2 < x < 2\} \underline{\hspace{1cm}} \{x \mid -4 < x \leq 4\};$ (3) $\{x \mid -2 < x < 2\} \underline{\hspace{1cm}} \{x \mid -4 < x \leq 4\};$

(4) $\emptyset \underline{\hspace{1cm}} \{x \mid x^3 = 8\};$

(5) $\emptyset \underline{\hspace{1cm}} \{x \mid x^2 + 1 = 0, x \in \mathbb{R}\};$

(6) $3 \underline{\hspace{1cm}} \{x \mid x < 3\};$

(7) $\{1, 3\} \underline{\hspace{1cm}} \{1, 2, 4\};$

(8) $\{x \mid 2 \leq x \leq 4\} \underline{\hspace{1cm}} \{x \mid 0 < x < 3\};$

(9) $\mathbb{Z}^+ \underline{\hspace{1cm}} \mathbb{N} \underline{\hspace{1cm}} \mathbb{Z} \underline{\hspace{1cm}} \mathbb{Q} \underline{\hspace{1cm}} \mathbb{R}.$

2. 下列关系正确的是 (). (Which of the following relationships is correct ().)

A. $2 \subseteq \{x \mid x > 0\}$ B. $\{2\} \in \{x \mid -2 < x < 4\}$ C. $3 \notin \{x \mid -3 \leq x < 3\}$ D. $\emptyset = \{0\}$

3. 已知集合 $M = \{2\}$, $N = \{0, 2, 4\}$, 则下列关系正确的是 (). (Given sets $M = \{2\}$ and

$N = \{0, 2, 4\}$, which of the following relationships is correct ().)

A. $M < N$ B. $M \in N$ C. $M \subseteq N$ D. $N \subseteq M$

4. 下列各组中的两个集合 M 和 N , 表示同一集合的是 (). (Which of the following pairs of sets M and N represent the same set ().)

A. $M = \{\pi\}$, $N = \{3.14\}$ B. $M = \{2, 3\}$, $N = \{(2, 3)\}$

C. $M = \{x \in \mathbb{N} \mid -1 < x \leq 1\}$, $N = \{1\}$ D. $M = \{1, 2, 3\}$, $N = \{3, 1, |-\sqrt{4}|\}$

5. 写出集合 $\{1, 3, 5, 7\}$ 的所有子集, 并指出哪些是它的真子集. (Write all subsets of the set

$\{1, 3, 5, 7\}$ and indicate which are its proper subsets.)



解：所有子集：____ (Solution: All subsets: ____)

真子集：____ (Proper subsets: ____)

习题 1.3 (Exercise 1.3)

1. 设集合 $A = \{-1, 1, 3, 5\}$, $B = \{-1, 2, 4, 6\}$, 则 $A \cup B =$ ____, $A \cap B =$ ____ .

Let sets $A = \{-1, 1, 3, 5\}$ and $B = \{-1, 2, 4, 6\}$, then $A \cup B =$ ____, $A \cap B =$ ____ .)

2. 设集合 $A = \{x \mid -3 < x \leq 4\}$, $B = \{x \mid -5 \leq x < 2\}$, 则 $A \cup B =$ ____, $A \cap B =$ ____ .

3. 设集合 $A = \{x \mid x \geq 0\}$, $B = \{x \mid x > 4\}$, 则 $A \cup B =$ ____, $A \cap B =$ ____ .

4. 设集合 $A = \{x \mid x < -1\}$, $B = \{x \mid x \leq 3\}$, 则 $A \cup B =$ ____, $A \cap B =$ ____ .

5. 设集合 $A = \{x \in \mathbb{R} \mid x^2 + 4 = 0\}$, $B = \{x \mid 2x - 4 = 0\}$, 则 $A \cup B =$ ____, $A \cap B =$ ____ .

6. 设集合 $U = \{0, 2, 4, 6, 8, 10\}$, $A = \{2, 6, 8\}$, $B = \{0, 4, 6, 10\}$, 则 $C_U A =$ ____, $C_U B =$ ____,

$(C_U A) \cap (C_U B) =$ ____ .

7. 设集合 $U = \{x \mid -5 \leq x < 5\}$, $A = \{x \mid -2 \leq x < 2\}$, 则 $C_U A =$ ____ .

8. 设集合 $U = \{x \in \mathbb{Z} \mid -5 \leq x \leq 5\}$, $A = \{x \in \mathbb{Z} \mid -2 \leq x < 2\}$, 则 $C_U A =$ ____ .

9. 设集合 $U = \mathbb{R}$, $A = (-4, 2)$, $B = [-1, 5]$, 则 $A \cup B =$ ____, $A \cap B =$ ____, $C_U A =$ ____,
 $C_U B =$ ____ .

10. 已知 $U = \{0, 1, 2, 3, 4, 5, 6, 7, 8\}$, $M = \{1, 2, 4, 5\}$, $N = \{0, 3, 5, 7\}$, 则 $C_U(M \cup N) =$ () . (Given

$U = \{0, 1, 2, 3, 4, 5, 6, 7, 8\}$, $M = \{1, 2, 4, 5\}$, $N = \{0, 3, 5, 7\}$, then $C_U(M \cup N) =$ () .)

A. $\{6, 8\}$ B. $\{5, 7\}$ C. $\{4, 6, 7\}$ D. $\{1, 3, 5, 6, 8\}$

11. 已知集合 $U = \{1, 2, 3, 4, 5, 6, 7\}$, $A = \{2, 3, 4, 5\}$, $B = \{2, 3, 6, 7\}$, 则 $B \cap (C_U A) =$ () . (Given

universal set $U = \{1, 2, 3, 4, 5, 6, 7\}$, $A = \{2, 3, 4, 5\}$, $B = \{2, 3, 6, 7\}$, then $B \cap (C_U A) =$ () .)

A. $\{1, 6\}$ B. $\{1, 7\}$ C. $\{6, 7\}$ D. $\{1, 6, 7\}$

自测题 (Self-test)

1. 用符号 “ $\in$ ” “ $\notin$ ” “ $\subseteq$ ” “ $\supseteq$ ” “ $\subset$ ” “ $\subsetneq$ ” 或 “ $=$ ” 填空 . (Fill in the blanks with the symbols

“ $\in$ ” “ $\notin$ ” “ $\subseteq$ ” “ $\supseteq$ ” “ $\subset$ ” “ $\subsetneq$ ” or “ $=$ ” .)

(1) 5 ____ $\{-4, 0, 4\}$;

(2) $\{0, 3\}$ ____ $\{-1, 0, 3, 5\}$;

(3) $\{x \mid 1 < x \leq 3\}$ ____ $\{x \mid -2 < x \leq 6\}$;

(4) $\emptyset$ ____ $\{x \mid 3x + 6 = 0\}$;

(5) 2 ____ $\{x \mid x < 2\}$;

(6) 0 ____ $\{x \mid x \leq 5\}$;

(7) $\sqrt{25}$ ____ $\mathbb{Q}$, $\frac{4}{7}$ ____ $\mathbb{Z}$, -5.2 ____ $\mathbb{R}$;



(8) $\{-1,0,1\} \underline{\hspace{1cm}} \{-1,2,3\}$.

2. 填空题 . (Fill in the blanks .)

(1) 设集合 $A = \{0,2,3,6\}$, $B = \{-1,0,4\}$, 则 $A \cup B = \underline{\hspace{1cm}}$, $A \cap B = \underline{\hspace{1cm}}$. ((1) Let sets $A = \{0,2,3,6\}$ and $B = \{-1,0,4\}$, then $A \cup B = \underline{\hspace{1cm}}$, $A \cap B = \underline{\hspace{1cm}}$.)

(2) 设集合 $A = \{x \mid -2 < x \leq 2\}$, $B = \{x \mid -4 \leq x < 1\}$, 则 $A \cup B = \underline{\hspace{1cm}}$, $A \cap B = \underline{\hspace{1cm}}$. (

(3) 设集合 $U = \{-1,0,2,3,7,8\}$, $A = \{0,2,8\}$, $B = \{2,7,8\}$, 则 $C_U A = \underline{\hspace{1cm}}$, $C_U B = \underline{\hspace{1cm}}$, $(C_U A) \cap (C_U B) = \underline{\hspace{1cm}}$.

(4) 设集合 $U = \{x \in \mathbb{Z} \mid -2 < x \leq 7\}$, $A = \{x \in \mathbb{Z} \mid -1 \leq x < 4\}$, 则 $C_U A = \underline{\hspace{1cm}}$.

(5) 设集合 $U = (0, +\infty)$, $A = [3,5]$, $B = (1,7]$, 则 $A \cup B = \underline{\hspace{1cm}}$, $A \cap B = \underline{\hspace{1cm}}$, $C_U A = \underline{\hspace{1cm}}$, $C_U B = \underline{\hspace{1cm}}$.

(6) 设全集 $U = \mathbb{R}$, 集合 $A = \{x \mid 0 \leq x \leq 2\}$, $B = \{y \mid 1 \leq y < 5\}$, 则 $(C_U A) \cap B = \underline{\hspace{1cm}}$. ((6) Let universal set $U = \mathbb{R}$, set $A = \{x \mid 0 \leq x \leq 2\}$, $B = \{y \mid 1 \leq y < 5\}$, then $(C_U A) \cap B = \underline{\hspace{1cm}}$.)

(7) 已知集合 $A = \{-2, -1, 0, 1\}$, $B = \{y \mid y = |x| - 1, x \in A\}$, 则 $A \cap B = \underline{\hspace{1cm}}$. ((7) Given set $A = \{-2, -1, 0, 1\}$ and $B = \{y \mid y = |x| - 1, x \in A\}$, then $A \cap B = \underline{\hspace{1cm}}$.)

(8) 已知全集 $U = \mathbb{R}$, $A = \{x \mid x \leq 0\}$, $B = \{x \mid x \geq 1\}$, 则集合 $C_U(A \cup B) =$ ____ . ((8)

Given universal set $U = \mathbb{R}$, $A = \{x \mid x \leq 0\}$, $B = \{x \mid x \geq 1\}$, then set $C_U(A \cup B) =$ ____ .)

(9) 若集合 $A = \{1, 2, 3, 5, 7\}$, $B = \{0, 2, 3, 4\}$, 则 $A \cap B$ 的子集个数为 ____ . ((9) If sets $A =$

$\{1, 2, 3, 5, 7\}$ and $B = \{0, 2, 3, 4\}$, then the number of subsets of $A \cap B$ is ____ .)